國立臺北科技大學
National Taipei University of Technology

資料結構
Data Structure

Lab 05

姓名:王顗程
學號:114310215

Lab01-Ex1. (題目編號)

Part 1: Understanding Basics

Refer to the class materials and Class06 PowerPoint slides to understand the fundamentals of Stack in C++.

Code

```
#include <iostream>
using namespace std;

struct Stack {
 int s[100];
 int top;
 Stack() { top = -1; }
};
Stack st;

void push(int n) {
 if (st.top == 99) {
 cout << "Stack 是滿的" << endl;
 }
 else {
 st.s[++st.top] = n;
 }
}

int pop() {
 if (st.top == -1) {
 cout << "Stack 是空的" << endl;
 return -1;
 }
 else {
 return st.s[st.top--];
 }
}
void printStack() {
 cout << "Stack 目前儲存元素有";  for (int i = 0;
```

```
i <= st.top; i++) {  cout << st.s[i] << " ";  }
 cout << endl;
}

int main()
{
 for (int i = 1; i <= 4; i++) {  push(i);
 cout << "push: " << i << endl;  printStack();
 }

 int x = pop();
 cout << "pop: " << x << endl;
printStack();
}
```

Lab01-Ex2. (題目編號)

Exercise: Shuffling and Dealing Cards (Use Stack) 1)

Shuffle a standard 52-card deck (excluding jokers). 2) Push the shuffled cards into a stack.

3) Draw cards from the stack and display the results.

To-do:

Code

```cpp
#include <iostream>
#include <vector>
#include <algorithm>
#include <random>
#include <string> // 記得引入 string
#include <stdexcept> // 用於 out_of_range 異常處理

using namespace std;

// 表示單張撲克牌的類別
class Card {
public:
    string colors;
    string rank;
    Card(string s, string r) : colors(s), rank(r) {}

    void display() const {
        cout << rank << " of " << colors << endl;
    }
};

// 實作 Stack
class Stack {
private:
    vector<Card> data; // 將變數名稱改為 data, 避免與類別名 Stack 衝突
public:
    void push(const Card& card) {
        data.push_back(card);
    }
    Card pop() {
        if (isEmpty()) {
            throw out_of_range("Stack is empty!");
        }
        Card topCard = data.back();
        data.pop_back();
        return topCard;
    }
    bool isEmpty() const {
        return data.empty();
    }
};

// 代表一副撲克牌的類別
class Deck {
private:
    vector<Card> cards;
    Stack shuffledDeck;

public: // *** 重要:建構子和函式必須設為 public, 外部才能呼叫 ***
    Deck() {
        string colors[] = { "Hearts", "Diamonds", "Clubs", "Spades" };
        string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9", "10", "J", "Q",
"K", "A" };

        for (int i = 0; i < 4; i++) {
```

```
            for (int j = 0; j < 13; j++) {
                cards.push_back(Card(colors[i], ranks[j]));
            }
        }
    }

    // 洗牌 (Hint: 使用 shuffle 函數)
    void shuffleDeck() {
        random_device rd;
        mt19937 g(rd());
        shuffle(cards.begin(), cards.end(), g);

        // 將洗好的牌存入 Stack
        for (const auto& card : cards) {
            shuffledDeck.push(card);
        }
    }

    // 發牌
    void drawAllCards() {
        int count = 1;
        while (!shuffledDeck.isEmpty()) {
            cout << count << ": ";
            Card c = shuffledDeck.pop();
            c.display();
            count++;
        }
    }
}; // *** 確保類別結尾有分號 ***

int main() {
    Deck deck; // 建立 deck 產生 52 張撲克牌
    deck.shuffleDeck(); // 進行洗牌並放入堆疊
    cout << "Shuffled deck:" << endl;
    deck.drawAllCards(); // 依序取出堆疊內的牌並顯示
    return 0;
}
```

Lab01-Ex3. (題目編號)	
Briefly describe the problem. (題目簡述)	
Code	
(a)	(b)
Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)
Discussion Section	

Lab01-Q1. (題目編號)
Briefly describe the problem. (題目簡述)
Code
Result

Microsoft Visual Studio 偵錯主控台

```
Shuffled deck:
1: 4 of Spades
2: 9 of Diamonds
3: 10 of Diamonds
4: 7 of Hearts
5: J of Clubs
6: 2 of Diamonds
7: J of Diamonds
8: A of Spades
9: 7 of Diamonds
10: 7 of Spades
11: 8 of Hearts
12: 8 of Diamonds
13: A of Diamonds
14: 6 of Clubs
15: Q of Spades
16: 8 of Clubs
17: 6 of Spades
18: 7 of Clubs
19: K of Spades
20: 2 of Clubs
21: 10 of Spades
22: 8 of Spades
23: 5 of Diamonds
24: A of Hearts
25: 5 of Hearts
26: 4 of Diamonds
27: 5 of Spades
28: 3 of Clubs
29: Q of Hearts
30: 10 of Clubs
31: Q of Diamonds
32: 5 of Clubs
33: 6 of Hearts
34: 3 of Diamonds
35: 2 of Spades
36: J of Hearts
37: J of Spades
38: Q of Clubs
39: A of Clubs
40: K of Diamonds
41: 6 of Diamonds
42: 9 of Spades
43: 3 of Spades
44: 4 of Clubs
45: 3 of Hearts
46: 10 of Hearts
47: 4 of Hearts
48: K of Hearts
49: K of Clubs
50: 2 of Hearts
51: 9 of Clubs
52: 9 of Hearts
```

檔案(F) 編輯(E) 檢視(V) Git(G) 專案(P) 建置(B) 偵錯(D) 測試(S) 分析(N) 工具(T) 延伸模組(X) 視窗(W) 說明(H) 搜尋 ConsoleApplication1 登入

Debug x64 本機 Windows 偵錯工具

ConsoleApplication1.cpp

ConsoleApplication1 Card

```cpp
// 實作 Stack
class Stack {
private:
    vector<Card> data; // 將變數名稱改為 data，避免與類別名 Stack 衝突
public:
    void push(const Card& card) {
        data.push_back(card);
    }
    Card pop() {
        if (isEmpty()) {
            throw out_of_range("Stack is empty!");
        }
        Card topCard = data.back();
        data.pop_back();
        return topCard;
    }
    bool isEmpty() const {
        return data.empty();
    }
};

// 代表一副撲克牌的類別
class Deck {
private:
    vector<Card> cards;
    Stack shuffledDeck;

public: // *** 重要：建構子和函式必須設為 public，外部才能呼叫 ***
    Deck() {
        string colors[] = { "Hearts", "Diamonds", "Clubs", "Spades" };
        string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9", "10", "J", "Q", "K", "A" };

        for (int i = 0; i < 4; i++) {
            for (int j = 0; j < 13; j++) {
                cards.push_back(Card(colors[i], ranks[j]));
            }
        }
    }

    // 洗牌 (Hint: 使用 shuffle 函數)
    void shuffleDeck() {
        random_device rd;
        mt19937 g(rd());
        shuffle(cards.begin(), cards.end(), g);

        // 將洗好的牌存入 Stack
        for (const auto& card : cards) {
            shuffledDeck.push(card);
        }
```

100 % 找不到任何問題

錯誤清單 輸出

就緒

Microsoft Visual Studio 偵錯主控台

```
Shuffled deck:
1: K of Diamonds
2: A of Diamonds
3: A of Spades
4: 3 of Hearts
5: 7 of Hearts
6: J of Diamonds
7: 9 of Spades
8: 5 of Spades
9: K of Clubs
10: A of Hearts
11: 6 of Diamonds
12: 7 of Clubs
13: Q of Hearts
14: 4 of Spades
15: 3 of Diamonds
16: Q of Diamonds
17: 4 of Hearts
18: 10 of Diamonds
19: 3 of Clubs
20: Q of Spades
21: A of Clubs
22: 6 of Clubs
23: J of Clubs
24: 8 of Hearts
25: 8 of Spades
26: 7 of Spades
27: 6 of Hearts
28: 5 of Hearts
29: 2 of Spades
30: 10 of Clubs
31: 8 of Diamonds
32: 4 of Diamonds
33: 10 of Hearts
34: 6 of Spades
35: 3 of Spades
36: 4 of Clubs
37: 5 of Clubs
38: K of Spades
39: 9 of Clubs
40: 9 of Diamonds
41: J of Spades
42: 10 of Spades
43: K of Hearts
44: 2 of Diamonds
45: 8 of Clubs
46: J of Hearts
47: 9 of Hearts
48: 2 of Hearts
49: Q of Clubs
50: 2 of Clubs
51: 5 of Diamonds
52: 7 of Diamonds

C:\Users\CNC\source\repos\ConsoleApplication1\x64\Debug\ConsoleApplication1.exe (流程 16364) 已結束，代碼為 0 (0x0)。
若要在偵錯停止時自動關閉主控台，請啟用 [工具] -> [選項] -> [偵錯] -> [偵錯停止時，自動關閉主控台]。
按任意鍵關閉此視窗…
```

Discussion

Microsoft Visual Studio 偵錯主控台

```
Shuffled deck:
1: 4 of Hearts
2: A of Spades
3: 4 of Clubs
4: 5 of Hearts
5: Q of Diamonds
6: 3 of Diamonds
7: J of Clubs
8: 5 of Clubs
9: 2 of Hearts
10: A of Clubs
11: 7 of Clubs
12: 3 of Spades
13: Q of Hearts
14: 9 of Hearts
15: K of Hearts
16: 10 of Clubs
17: 8 of Clubs
18: 5 of Diamonds
19: 10 of Diamonds
20: 4 of Diamonds
21: K of Clubs
22: J of Diamonds
23: J of Hearts
24: 2 of Spades
25: Q of Clubs
26: 2 of Clubs
27: J of Spades
28: 7 of Diamonds
29: 10 of Hearts
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43: 3 of Clubs
44: Q of Spades
45: 6 of Diamonds
46: 6 of Clubs
47: 9 of Spades
48: 8 of Hearts
49: 9 of Clubs
50: K of Diamonds
51: A of Hearts
52: 2 of Diamonds
```

Lab01-Q#. (題目編號)		
Briefly describe the problem. (題目簡述)		
Code		
Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	
result		
(a)	(b)	(c)
	Show your result in tables or screenshots. (請使用表格或是截圖呈現結果。)	
Discussion		
Discuss and conclude your results, or answer questions. (實驗討論、結論或回答問題。)		

[EXAMPLE] Q1. Factorial Function

The factorial function n! has value 1 when n<=1 and value n*(n-1)! when n>1.

Please write both a recursive and an iterative C function. Indicate the input, output, and give proper comments. Test your codes and log the results.

Code

<iterative version>

```c
#include<stdio.h>
#include <stdlib.h>
int main()
{
int n,answer;
char i;
printf("Factorial function n!,n=");
scanf("%d",&n); //輸入要查詢的 n
if(n<=1)
{
printf("Answer=1"); //小於 1 時答案為 1
}
else //大於 1 時
{
answer=n; //計算 (n>12 時超出 int 儲存上限)
for(i=(n-1);i>1;i--)
{
answer=answer*i;
}
printf("Answer=%d\n",answer);
}
system("pause");
return 0;
}
```

<recursive version>

…

result

```
Factorial function n!,n=7
Answer=5040
```

Discussion